Lecturers: Thomas Quella and Nick Beaton
School of Mathematics and Statistics

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University of Melbourne

MAST20030: Differential Equations
Tutorial Solutions: Week 5

Solution to Question 1.

(a) This integral converges by direct calculation:

$$\int_0^\infty \frac{1}{1+t^2}\,\mathrm{d}t = \lim_{s\to\infty}\int_0^s \frac{1}{1+t^2}\,\mathrm{d}t = \lim_{s\to\infty}\tan^{-1}s = \frac{\pi}{2}.$$

(b) This integral diverges by substituting $u = t^2$:

$$\int_0^\infty \frac{t}{1+t^2}\,\mathrm{d}t = \lim_{s\to\infty}\int_0^{s^2}\frac{1}{2}\frac{1}{1+u}\,\mathrm{d}u = \lim_{s\to\infty}\log(1+s^2) = \infty.$$

(c) This integral converges by a comparison test:

$$|\cos(\mathrm{e}^{t^2})| \leqslant 1 \;\Rightarrow\; \left|\mathrm{e}^{-t}\cos(\mathrm{e}^{t^2})\right| \leqslant \mathrm{e}^{-t} \;\Rightarrow\; \left|\int_0^\infty \mathrm{e}^{-t}\cos(\mathrm{e}^{t^2})\,\mathrm{d}t\right| \leqslant \int_0^\infty \mathrm{e}^{-t}\,\mathrm{d}t < \infty.$$

Solution to Question 2.

(a) The integral definition of the Laplace transform of t^2 gives

$$\mathscr{L}\{t^2\} = \int_0^\infty t^2\mathrm{e}^{-st}\mathrm{d}t$$

Using integration by parts we have

$$\begin{aligned}\int_0^\infty t^2\mathrm{e}^{-st}\mathrm{d}t &= \left[-\frac{t^2}{s}\mathrm{e}^{-st}\right]_0^\infty - \int_0^\infty -\frac{2t}{s}\mathrm{e}^{-st}\mathrm{d}t = \int_0^\infty \frac{2t}{s}\mathrm{e}^{-st}\mathrm{d}t\\ &= \left[-\frac{2t}{s^2}\mathrm{e}^{-st}\right]_0^\infty - \int_0^\infty -\frac{2}{s^2}\mathrm{e}^{-st}\mathrm{d}t = \int_0^\infty \frac{2}{s^2}\mathrm{e}^{-st}\mathrm{d}t\\ &= \left[-\frac{2}{s^3}\mathrm{e}^{-st}\right]_0^\infty = \frac{2}{s^3}.\end{aligned}$$

This Laplace transform exists for $\mathrm{Re}(s) > 0$.

(b) By linearity of the Laplace transform we have

$$\mathscr{L}\{f\} = 3\mathscr{L}\{\mathrm{e}^{-4t}\} + \mathscr{L}\left\{\sinh\frac{t}{2}\right\} + \frac{1}{2}\mathscr{L}\{t^5\}$$

We may then evaluate each term from tables giving

$$\mathscr{L}\{f\} = \frac{3}{s+4} + \frac{1/2}{s^2-1/4} + \frac{5!}{2s^6} = \frac{3}{s+4} + \frac{2}{4s^2-1} + \frac{60}{s^6}, \qquad \mathrm{Re}(s) > \frac{1}{2}$$

Solution to Question 3.

(a) $\mathscr{L}\{(t+7)^2\} = \mathscr{L}\{(t^2+14t+49)\} = \dfrac{2}{s^3} + \dfrac{14}{s^2} + \dfrac{49}{s}$, for all $\mathrm{Re}(s) > 0$. Thus,

$$\begin{aligned}\mathscr{L}\{(t+7)^2\cosh t\} &= \frac{1}{2}\mathscr{L}\{(t+7)^2(\mathrm{e}^t + \mathrm{e}^{-t})\} \\ &= \frac{1}{(s-1)^3} + \frac{7}{(s-1)^2} + \frac{49/2}{s-1} + \frac{1}{(s+1)^3} + \frac{7}{(s+1)^2} + \frac{49/2}{s+1}, \qquad \mathrm{Re}(s) > 1.\end{aligned}$$

(b) The Laplace transform of $\mathrm{e}^{(t-1)^2}$ does not exist as $\displaystyle\int_0^\infty \mathrm{e}^{-st}\mathrm{e}^{(t-1)^2}\,\mathrm{d}t$ diverges. Why? Because the integrand diverges to ∞ as $t \to \infty$, hence the "area under the curve" is infinite, i.e., divergent.

(c) Writing $\cos(\omega t + \theta) = \cos(\omega t)\cos\theta - \sin(\omega t)\sin\theta$, we get

$$\mathscr{L}\{\cos(\omega t+\theta)\} = \mathscr{L}\{\cos(\omega t)\}\cos\theta - \mathscr{L}\{\sin(\omega t)\}\sin\theta = \frac{s\cos\theta - \omega\sin\theta}{s^2+\omega^2}, \qquad \mathrm{Re}(s) > 0.$$

Solution to Question 4.

(a) $\mathscr{L}^{-1}\left\{\dfrac{2s+7}{s^2+16}\right\} = 2\mathscr{L}^{-1}\left\{\dfrac{s}{s^2+4^2}\right\} + \dfrac{7}{4}\mathscr{L}^{-1}\left\{\dfrac{4}{s^2+4^2}\right\} = 2\cos(4t) + \dfrac{7}{4}\sin(4t)$.

(b) This time we use partial fractions and the cover method: $\dfrac{12}{s^2-2s-3} = \dfrac{12}{(s-3)(s+1)} = \dfrac{3}{s-3} - \dfrac{3}{s+1}$. Thus, $\mathscr{L}^{-1}\left\{\dfrac{12}{s^2-2s-3}\right\} = 3(\mathrm{e}^{3t} - \mathrm{e}^{-t})$.

(c) We again use partial fractions, but this time the cover method only gives us the coefficient of the linear factor: $\dfrac{s^2+9s-2}{(s^2+16)(s-2)} = \dfrac{As+B}{s^2+16} + \dfrac{1}{s-2}$. Substituting $s = 0$ gives $B = 9$ and then it's easy to check that $A = 0$. Thus,

$$\mathscr{L}^{-1}\left\{\frac{s^2+5s-3}{(s^2+16)(s-2)}\right\} = \mathscr{L}^{-1}\left\{\left[\frac{9}{s^2+16} + \frac{1}{s-2}\right]\right\} = \frac{9}{4}\sin(4t) + \mathrm{e}^{2t}.$$

(d) Using the s-shift theorem, $\mathscr{L}^{-1}\left\{\dfrac{5}{(s-3)^4}\right\} = \dfrac{5}{6}\mathscr{L}^{-1}\left\{\dfrac{3!}{(s-3)^4}\right\} = \dfrac{5}{6}\mathrm{e}^{3t}\mathscr{L}^{-1}\left\{\dfrac{3!}{s^4}\right\} = \dfrac{5}{6}t^3\mathrm{e}^{3t}$

(e) $\dfrac{4s^2-s+10}{s^2(s+1)} = \dfrac{A}{s+1} + \dfrac{B}{s} + \dfrac{C}{s^2} = \dfrac{As^2+Bs^2+Bs+Cs+C}{s^2(s+1)}$. Equating coefficients gives $A+B = 4$, $B+C = -1$ and $C = 10$. Hence $B = -11$ and $A = 15$. Thus

$$\mathscr{L}^{-1}\left\{\frac{4s^2-s+10}{s^2(s+1)}\right\} = \mathscr{L}^{-1}\left\{\frac{15}{s+1}\right\} - \mathscr{L}^{-1}\left\{\frac{11}{s}\right\} + \mathscr{L}^{-1}\left\{\frac{10}{s^2}\right\} = 15\mathrm{e}^{-t} - 11 + 10t$$

(f) Using the s-shift theorem, $\mathscr{L}^{-1}\left\{\dfrac{s}{(s+4)^4}\right\} = \mathrm{e}^{-4t}\mathscr{L}^{-1}\left\{\dfrac{s-4}{s^4}\right\} = \mathrm{e}^{-4t}\left\{\dfrac{t^2}{2} - \dfrac{2t^3}{3}\right\}$